Internet Access for Iran

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Version: 1.0

Date: 20 March 2026

Abstract:
Provide fast, reliable internet access across Iran, including urban, rural, and remote areas. The plan combines fibre deployment in major cities, Wi-Fi hotspots and mobile gateways for hard-to-reach areas, upgraded cellular coverage, and satellite internet options. This initiative will bridge the digital divide, enable economic growth, create jobs, and prepare the country for a connected future.

Why It's Interesting:

- Iran is a large country with uneven internet coverage, limiting economic and social opportunities.
- Immediate need for connectivity in remote areas to support education, commerce, and communication.
- Potential to develop domestic internet technologies and attract international investment.
- Creates large-scale employment in engineering, installation, field support, and education.

Approach:

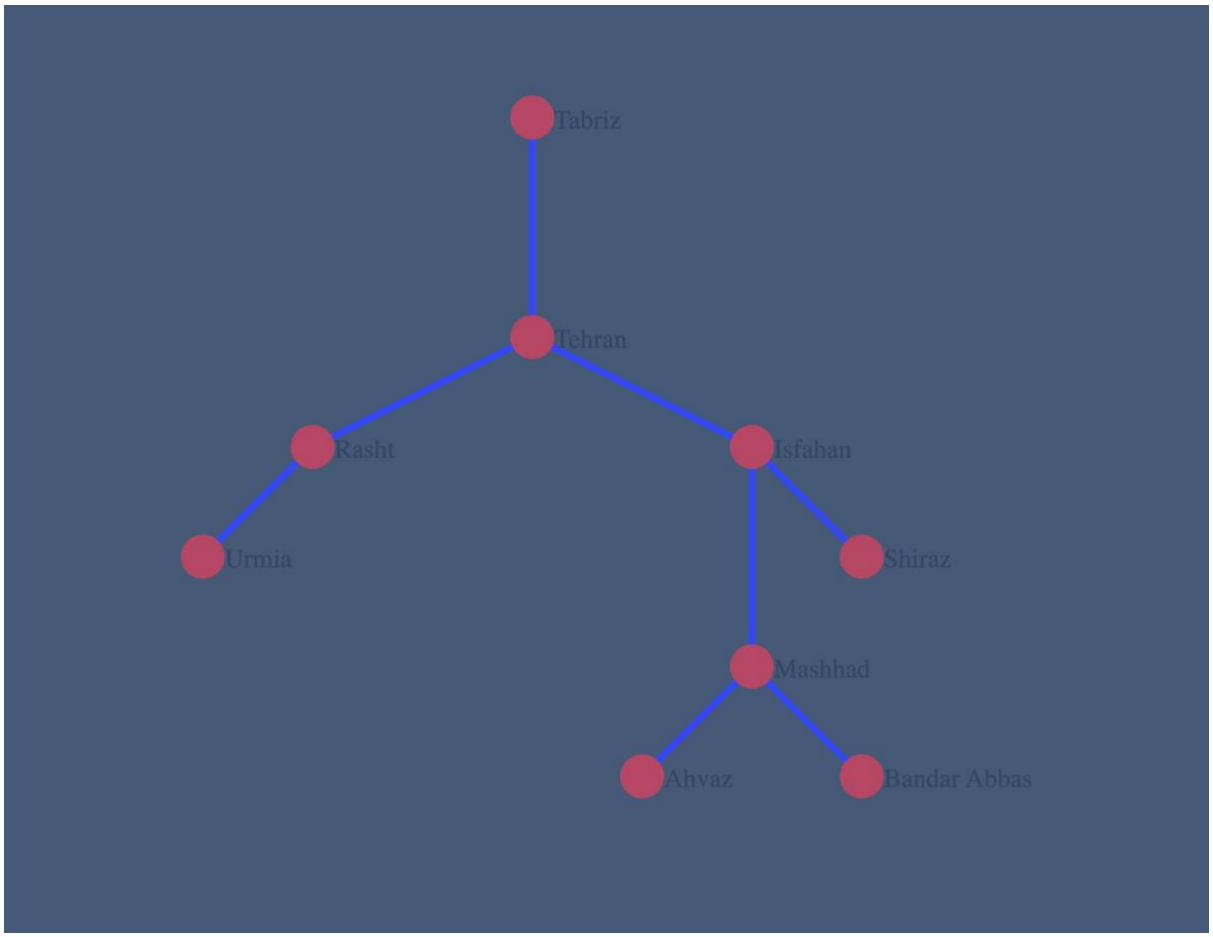
1. Deploy fibre networks in major cities (e.g., Tehran, Mashhad) and build mid-size exchange/Points of Interconnect (POIs).
2. Provide mobile gateways ("Wi-Fi on wheels") for remote and hard-to-reach areas until fibre is available. These can be equipped with strong receivers.
3. Upgrade cellular base stations in dense areas for improved bandwidth and coverage.
4. Use Fibre-to-the-Node (FTTN) technology in areas with existing copper lines, gradually transitioning to fibre-to-the-premises (FTTP).
5. Promote satellite internet (e.g., Starlink) and encourage competition through international partnerships and domestic assembly/production of related technologies.
6. Timeline: 1–2 years for city-wide fibre and infrastructure upgrades, with Wi-Fi hotspots deployed in parallel.
7. Methods & Tools: DWDM links between cities, OTDR testing for fibre, redundant POIs for reliability, scalable cellular upgrades.

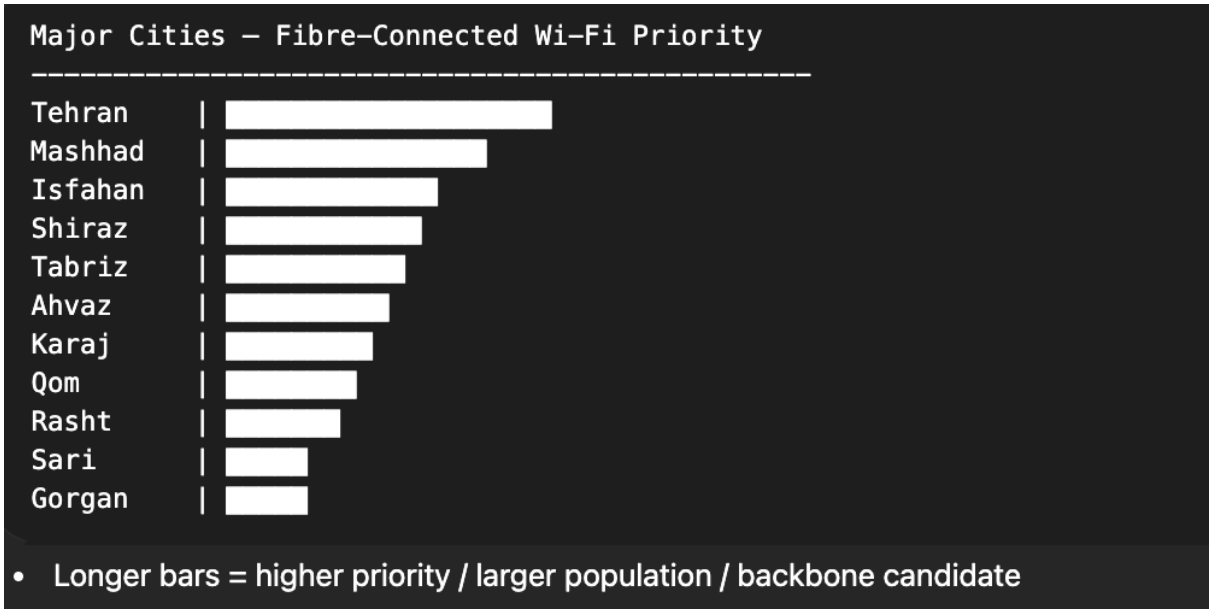
Help Needed:

- Investment and partnerships from large telecom companies (Telstra, Vodafone, Optus, Amazon, Microsoft, etc) and global equipment providers (Siemens, Nokia, Coriant, Motorola).
- Local and international network design engineers for planning and implementation.
- Government approvals, budget allocation, and coordination with municipal planning.
- Technical talent for deployment, maintenance, and support, as well as training programs for engineers and field technicians.

Supporting Material:

- Maps of proposed fibre routes and POI locations.
- Diagrams of mobile gateway setup and FTTN/FTTP transition.
- References to DWDM, OTDR, and cellular upgrade technologies.





Summary Approach

1. **Phase 1 – Fibre Backbone:**
   - Tehran, Mashhad, Isfahan, Shiraz, Tabriz, Ahvaz, Karaj, Qom.
2. **Phase 2 – Fibre Extension:**
   - Secondary cities: Rasht, Sari, Gorgan, Yazd, Kerman.
3. **Satellite / Mobile Wi-Fi Hotspots:**
   - Zahedan, Bam, rural mountain regions, border towns.
4. **Redundancy & Reliability:**
   - Multiple POIs in large cities
   - Satellite backup for remote/fibre-limited areas